

Relative symlink pitfalls

Symlinks let a project point at shared IP without copying files

Relative to the link, not your cwd

- The target string is resolved from the directory that contains the symlink
- From project slash external, shared lib is up two levels then shared slash lib
- Prefer relative inside a repo you will move; prefer absolute only for fixed system tools
- Moving only the symlink to a new folder often breaks a relative link

Browser lab

INTERACTIVE TOOL

Relative symlink pitfalls

Relative targets survive a whole-tree move; absolute targets often don't. Move the link alone (or the target alone) and watch relative links break.

Starter example: Under `/home/lab/chip/links/to_top_rel -> ../rtl/top.v` (relative) and `to_top_abs` (absolute). Relocate the whole tree – relative stays good; move only the link – relative breaks.

Load starter example

Challenges 0 / 22 cleared

Quiz: base: Relative symlink targets resolve against the? Answer: link dir

Answer

Show hintCheckNextIdle

Quiz: base Starter both OK readlink rel Tree move: abs OK Tree move: abs break Quiz: tree move

Move link breaks rel Move abs OK Quiz: link move Move target Quiz: abs use Quiz: rel use Rel string

Abs prefix Sibling rel opt root realpath starter Quiz: not end Count broken Quiz: target move Reset

Quiz: rule

Relative vs absolute

Digital Design and Verification Platform

HomeTools

HomeToolsRelative links

Move scenarios

Project root: /home/lab/chip

Reset starter tree

Move whole tree ~ /opt/chip

Move only to_top_rel ~ project root

Move only rtl/top.v ~ rtl/old/top.v

Move only to_top_abs ~ project root

Starts: compare to_top_rel vs to_top_abs, then try move scenario

lab\$ readlink . realpath . ls - help

readlink to_top_rel readlink to_top_abs realpath to_top_rel

ls links

Links status

link	stored	target	status
./links/to_top_abs	/home/lab/chip/rtl/top.v	OK ~	/home/lab/chip/rtl/top.v
./links/to_top_rel	../rtl/top.v	OK ~	/home/lab/chip/rtl/top.v
./rtl/alu_link	alu.v	OK ~	/home/lab/chip/rtl/alu.v

Tree

```
/home/lab/chip/  
Makefile  
links/  
  to_top_abs -> /home/lab/chip/rtl/top.v  
  to_top_rel -> ../rtl/top.v  
rtl/  
  alu.v  
  alu_link -> alu.v  
  top.v  
tb/  
  tb_top.v
```

Cheat sheet

Situation	Relative	Absolute
Move whole tree to new prefix	OK	Often broken
Move only the symlink	Often broken	OK if target stays
Move only the target file	Broken	Broken

Relative target `../rtl/top.v` is joined to the link's parent dir – not your cwd.

Prefer relative links inside a repo you will relocate; prefer absolute for fixed system tools.

Link relative lab

Real shell practice

```
wsl - module22-link-relative/examples/link_relative

# Track A - real Unix
# real shell session (Track A)

$ ls -la external/
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu   0 Jul 18 16:34 .gitkeep
lrwxrwxrwx 1 yongfu yongfu  18 Jul 18 16:34 lib.v → ../../shared/lib.v

$ readlink external/lib.v
../../shared/lib.v

$ cat external/lib.v
// Shared library for link_relative example
module lib;
endmodule

$ ln -s "${realpath ../../shared/lib.v}" external/abs_lib.v

$ ls -la external/
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 2026 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ..
-rwxrwxrwx 1 yongfu yongfu   0 Jul 18 16:34 .gitkeep
lrwxrwxrwx 1 yongfu yongfu 106 Jul 20 2026 abs_lib.v → /mnt/d/proj/designs/learning/courses/learn_unix/module22-link-
relative/examples/link_relative/shared/lib.v
lrwxrwxrwx 1 yongfu yongfu  18 Jul 18 16:34 lib.v → ../../shared/lib.v

$ readlink external/abs_lib.v
/mnt/d/proj/designs/learning/courses/learn_unix/module22-link-relative/examples/link_relative/shared/lib.v

$ rm external/abs_lib.v
```

Real shell — relative link to shared IP

Real shell practice — try these

```
# ls -la external/ — see lib.v -> ../../shared/lib.v
ls -la external/
```

```
# readlink external/lib.v — print the stored relative target
readlink external/lib.v
```

```
# cat external/lib.v — follow the link and show shared content
cat external/lib.v
```

```
# ln -s "$(realpath ...)" — absolute contrast (fixed path; breaks if
tree
moves)
```

```
ln -s "$(realpath ../../shared/lib.v)" external/abs_lib.v
```

```
readlink external/abs_lib.v
```

Pitfalls to watch

- Count the dots from the link's directory, not from the repo root by habit
- Do not move a relative symlink alone without updating its target string
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run whole-tree relocate and move-only-link
- On the real shell, inspect the external lib link with readlink and cat
- When you are ready